INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Atelier Informatique TD_2

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Billy Rolph EXUME

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Niveau : L3

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L’objectif de ce TD est de :

1. Travailler en équipe avec Git et Github
2. Gérer les branches proprement
3. Utiliser les Pull Requests
4. Résoudre des conflits
5. Historique Gi

Sur Github :

On a créé un dépôt nommé cafe-associatif

1) Mise à jour de la branche main locale après la fusion des Pull Requests

```
MINGW64:/c/Users/rbill/cafe-associatif

rbill@ROLPH MINGW64 ~/cafe-associatif (feature/accueil)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$ git pull origin main
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (10/10), done.
remote: Total 11 (delta 3), reused 5 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (11/11), 4.24 KiB | 131.00 KiB/s, done.
From https://github.com/Rbill1y/cafe-associatif
 * branch            main       -> FETCH_HEAD
   4364b2e..d967086  main       -> origin/main
Updating 4364b2e..d967086
Fast-forward
 README.md    | 42 ++++++++++++++++++++++++++++++++++++++++++
 contact.html | 42 ++++++++++++++++++++++++++++++++++++++++++
 index.html   | 19 +++++++++++++++++---
 menu.html    | 40 ++++++++++++++++++++++++++++++++++++++++
 4 files changed, 140 insertions(+), 3 deletions(-)
 create mode 100644 contact.html
 create mode 100644 menu.html

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$ |
```

2) Création des fichiers de base et premier commit/push vers GitHub

```
MINGW64:/c/Users/rbill/cafe-associatif

rbill@ROLPH MINGW64 ~
$ cd cafe-associatif

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$ touch index.html css/style.css README.md

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$ git add .

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$ git commit -m "initialisation du projet et structure de Base"
[main (root-commit) 4364b2e] initialisation du projet et structure de Base
 3 files changed, 64 insertions(+)
 create mode 100644 README.md
 create mode 100644 css/style.css
 create mode 100644 index.html

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$ git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 1.01 KiB | 1.01 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Rbil1y/cafe-associatif.git
 * [new branch]      main -> main

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$
```

```
MINGW64:/c/Users/rbill/cafe-associatif
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Rbil1y/cafe-associatif.git
 * [new branch]      main -> main

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$ git checkout -b feature/accueil
Switched to a new branch 'feature/accueil'

rbill@ROLPH MINGW64 ~/cafe-associatif (feature/accueil)
$
```

3) Création de la branche feature/accueil après le push initial

- **Commit et push de la branche feature/accueil vers GitHub**

```
MINGW64:/c/Users/rbill/cafe-associatif

rbill@ROLPH MINGW64 ~/cafe-associatif (main)
$ git checkout -b feature/accueil
Switched to a new branch 'feature/accueil'

rbill@ROLPH MINGW64 ~/cafe-associatif (feature/accueil)
$ git add .

rbill@ROLPH MINGW64 ~/cafe-associatif (feature/accueil)
$ git commit -m "Amélioration de l'accueil et ajout du README.md"
[feature/accueil dba8b15] Amélioration de l'accueil et ajout du README.md
 2 files changed, 58 insertions(+), 3 deletions(-)

rbill@ROLPH MINGW64 ~/cafe-associatif (feature/accueil)
$ git push origin feature/accueil
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.68 KiB | 1.68 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/accueil' on GitHub by visiting:
remote:      https://github.com/Rbil1y/cafe-associatif/pull/new/feature/accueil
remote:
To https://github.com/Rbil1y/cafe-associatif.git
 * [new branch]      feature/accueil -> feature/accueil

rbill@ROLPH MINGW64 ~/cafe-associatif (feature/accueil)
$
```

4) Clone du dépôt par Ruth-Jose (Jose) et création de la branche feature/menu

MINGW64:/c/Users/keyru/cafe-associatif

```
keyru@jose MINGW64 ~ (main)
$ git clone https://github.com/Rbil1y/cafe-associatif.git
Cloning into 'cafe-associatif'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.

keyru@jose MINGW64 ~ (main)
$ cd cafe-associatif

keyru@jose MINGW64 ~/cafe-associatif (main)
$ git checkout -b feature/menu
Switched to a new branch 'feature/menu'

keyru@jose MINGW64 ~/cafe-associatif (feature/menu)
$ |
```

MINGW64:/c/Users/keyru/cafe-associatif

```
keyru@jose MINGW64 ~/cafe-associatif (feature/menu)
$ git add .
warning: in the working copy of 'menu.html', LF will be replaced by CRLF the nex
t time Git touches it

keyru@jose MINGW64 ~/cafe-associatif (feature/menu)
$ git commit -m "Ajout de la page menu et de ses styles"
[feature/menu 6f0e2a1] Ajout de la page menu et de ses styles
 1 file changed, 40 insertions(+)
 create mode 100644 menu.html

keyru@jose MINGW64 ~/cafe-associatif (feature/menu)
$ git push origin feature/menu
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 787 bytes | 787.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/menu' on GitHub by visiting:
remote:      https://github.com/Rbil1y/cafe-associatif/pull/new/feature/menu
remote:
To https://github.com/Rbil1y/cafe-associatif.git
 * [new branch]      feature/menu -> feature/menu

keyru@jose MINGW64 ~/cafe-associatif (feature/menu)
$
```

5) **Commit et push de la page menu.html sur la branche feature/menu**

```
MINGW64:/c/Users/keyru/cafe-associatif

keyru@jose MINGW64 ~/cafe-associatif (feature/menu)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

keyru@jose MINGW64 ~/cafe-associatif (main)
$ git pull origin main
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 12 (delta 2), reused 6 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (12/12), 5.28 KiB | 122.00 KiB/s, done.
From https://github.com/Rbil1y/cafe-associatif
 * branch            main       -> FETCH_HEAD
   4364b2e..d967086  main       -> origin/main
Updating 4364b2e..d967086
Fast-forward
 README.md    | 42 ++++++++++++++++++++++++++++++++++++++++++++
 contact.html | 42 ++++++++++++++++++++++++++++++++++++++++++++
 index.html   | 19 +++++++++++++++++---
 menu.html    | 40 ++++++++++++++++++++++++++++++++++++++++
 4 files changed, 140 insertions(+), 3 deletions(-)
 create mode 100644 contact.html
 create mode 100644 menu.html

keyru@jose MINGW64 ~/cafe-associatif (main)
$
```

6) Retour sur main et mise à jour locale après fusion des PR (Jose)

```
MINGW64:/c/Users/keyru/cafe-associatif

keyru@jose MINGW64 ~/cafe-associatif (main)
$ git checkout -b feature/nav-update
Switched to a new branch 'feature/nav-update'

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git add .

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git commit -m "Amelioration navigation avec flexbox"
On branch feature/nav-update
nothing to commit, working tree clean

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git push origin faeture/nav-update
error: src refspec faeture/nav-update does not match any
error: failed to push some refs to 'https://github.com/Rbil1y/cafe-associatif.gi
t'

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git push origin feature/nav-update
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/nav-update' on GitHub by visiting:
remote:      https://github.com/Rbil1y/cafe-associatif/pull/new/feature/nav-upda
te
remote:
To https://github.com/Rbil1y/cafe-associatif.git
 * [new branch]      feature/nav-update -> feature/nav-update

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$
```

7) **Création de la branche feature/nav-update et push (correction faute de frappe incluse)**

```
MINGW64:/c/Users/keyru/cafe-associatif
$ git add css/style.css

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git commit -m "Amelioration navigation avec flexbox"
On branch feature/nav-update
nothing to commit, working tree clean

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git push origin feature/nav-update
Everything up-to-date

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git add css/style.css

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git commit -m "Amelioration navigation avec flexbox"
[feature/nav-update c966c85] Amelioration navigation avec flexbox
 1 file changed, 11 insertions(+), 6 deletions(-)

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$ git push origin feature/nav-update
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 871 bytes | 871.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Rbil1y/cafe-associatif.git
   d967086..c966c85  feature/nav-update -> feature/nav-update

keyru@jose MINGW64 ~/cafe-associatif (feature/nav-update)
$
```

8) Clone du dépôt par Brian et création de la branche feature/contact

```
MINGW64:/c/Users/Brian ANTOINE/cafe-associatif

Brian ANTOINE@Bri MINGW64 ~ (master)
$ git clone https://github.com/Rbil1y/cafe-associatif.git
Cloning into 'cafe-associatif'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.

Brian ANTOINE@Bri MINGW64 ~ (master)
$ cd cafe-associatif

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (main)
$ git checkout -b feature/contact
Switched to a new branch 'feature/contact'

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/contact)
$
```

9) Commit et push de la page contact.html sur la branche feature/contact

```
MINGW64:/c/Users/Brian ANTOINE/cafe-associatif
Switched to a new branch 'feature/contact'

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/contact)
$ git add .
warning: in the working copy of 'contact.html', LF will be replaced by CRLF the
next time Git touches it

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/contact)
$ git commit -m "Ajout  de la page contact et de ses styles"
[feature/contact 4fb2bf3] Ajout  de la page contact et de ses styles
 1 file changed, 42 insertions(+)
 create mode 100644 contact.html

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/contact)
$ git push origin feature/contact
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 897 bytes | 448.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/contact' on GitHub by visiting:
remote:      https://github.com/Rbil1y/cafe-associatif/pull/new/feature/contact
remote:
To https://github.com/Rbil1y/cafe-associatif.git
 * [new branch]      feature/contact -> feature/contact

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/contact)
$
```

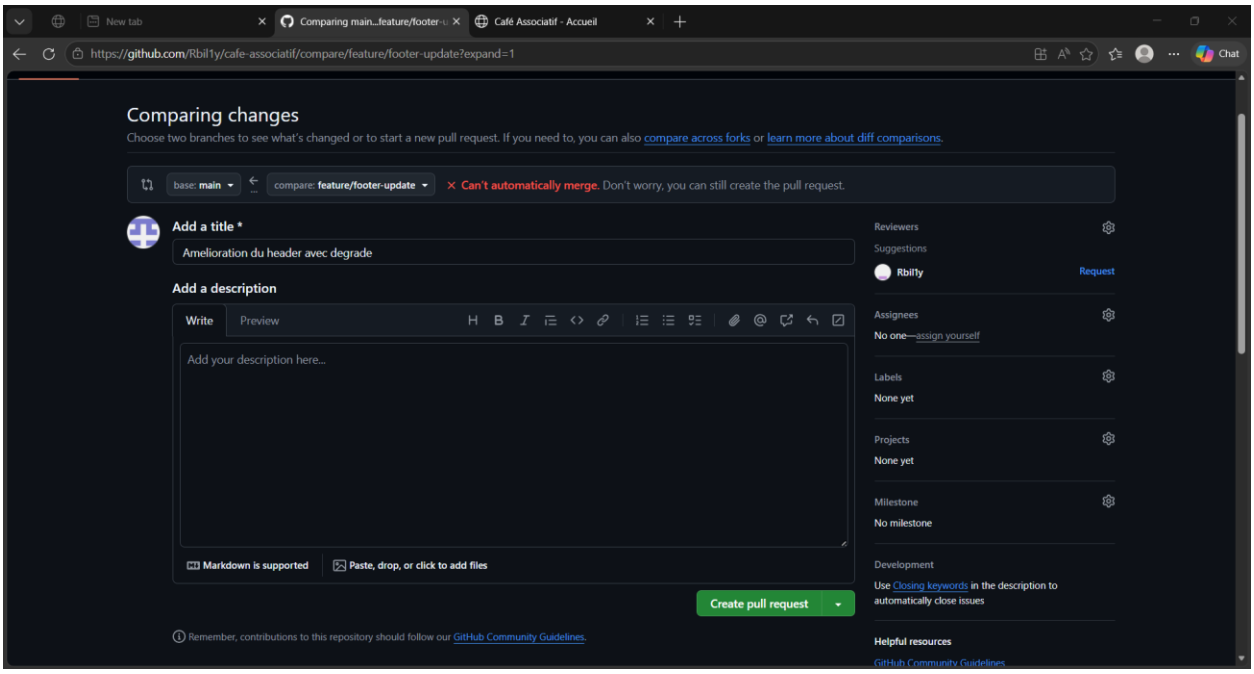
10) Mise à jour de la branche main de Brian après fusion des PR (contact + menu)

```
MINGW64:/c/Users/Brian ANTOINE/cafe-associatif

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/contact)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (main)
$ git pull origin main
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 12 (delta 3), reused 6 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (12/12), 5.17 KiB | 26.00 KiB/s, done.
From https://github.com/Rbil1y/cafe-associatif
 * branch            main       -> FETCH_HEAD
   4364b2e..d967086  main       -> origin/main
Updating 4364b2e..d967086
Fast-forward
 README.md    | 42 ++++++++++++++++++++++++++++++++++++++++++
 contact.html | 42 ++++++++++++++++++++++++++++++++++++++++++
 index.html   | 19 +++++++++++++++++---
 menu.html    | 40 ++++++++++++++++++++++++++++++++++++++++
 4 files changed, 140 insertions(+), 3 deletions(-)
 create mode 100644 contact.html
 create mode 100644 menu.html

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (main)
$
```

11) **Commit et push des modifications CSS du footer/header sur la branche feature/footer-update**

```
MINGW64:/c/Users/Brian ANTOINE/cafe-associatif
Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$ git commit -m "Amelioration du header avec degrade"
On branch feature/footer-update
nothing to commit, working tree clean

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$ git push origin feature/footer-update
Everything up-to-date

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$ git add css/style.css

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$ git commit -m "Amelioration du header avec degrade"
[feature/footer-update defe888] Amelioration du header avec degrade
 1 file changed, 10 insertions(+), 1 deletion(-)

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$ git push origin feature/footer-update
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 688 bytes | 344.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Rbil1y/cafe-associatif.git
   d967086..defe888  feature/footer-update -> feature/footer-update

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$
```

12) Tentative de création d'une Pull Request sur GitHub : détection automatique d'un conflit avec main



```
MINGW64:/c/Users/Brian ANTOINE/cafe-associatif
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (main)
$ git pull origin main
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 5 (delta 2), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.59 KiB | 26.00 KiB/s, done.
From https://github.com/Rbil1y/cafe-associatif
 * branch            main       -> FETCH_HEAD
   d967086..bb5fcc3  main       -> origin/main
Updating d967086..bb5fcc3
Fast-forward
 css/style.css | 17 +++++++++++------
 1 file changed, 11 insertions(+), 6 deletions(-)

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (main)
$ git checkout feature/footer-update
Switched to branch 'feature/footer-update'

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$ git merge main
Auto-merging css/style.css
CONFLICT (content): Merge conflict in css/style.css
Automatic merge failed; fix conflicts and then commit the result.

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update|MERGING)
$
```

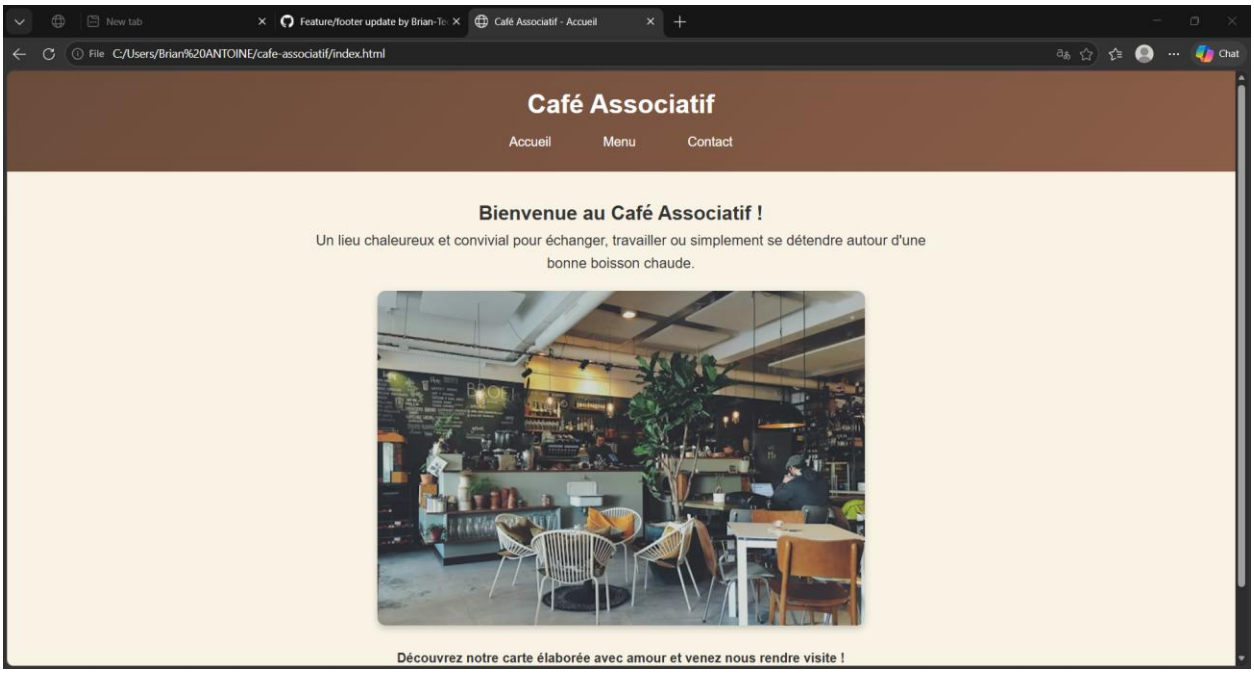
- ✓ **Après résolution manuelle du conflit : ajout et commit du fichier corrigé**

```
C: > Users > Brian ANTOINE > cafe-associatif > css > style.css > ...
26  main {
28      margin: 2rem auto;
29      padding: 0 1rem;
30      min-height: 60vh; /* Garde le footer en bas */
31  }
32
33  footer {
34      background-color: #6b4c3b;
35      color: white;
36      text-align: center;
37      padding: 1rem;
38      margin-top: 2rem;
39  }
    Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
40  <<<<<<< HEAD (Current Change)
41
42  /* Styles ajoutés par Brian (Contact) */
43  .contact-container { margin-top: 1.5rem; }
44  .form-group { margin-bottom: 1rem; }
45  .form-group label { display: block; font-weight: bold; color: #6b4c3b; }
46  .form-group input, .form-group textarea { width: 100%; padding: 0.5rem; border: 1px solid #ddd; }
47  .btn-submit { background-color: #6b4c3b; color: white; padding: 0.75rem 1.5rem; border: none; cursor: pointer; }
48  =======
49  /* Styles ajoutés par Ruth (Menu) */
50  .menu-section { margin-bottom: 2rem; }
51  .menu-section h3 { color: #6b4c3b; border-bottom: 2px solid #6b4c3b; margin-bottom: 1rem; }
52  .menu-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); gap: 1.5rem; }
53  .card { background: white; border-radius: 8px; padding: 1rem; text-align: center; box-shadow: 0 2px 5px rgba(0,0,0,0.1); }
54
55  >>>>>>> main (Incoming Change)
56
```

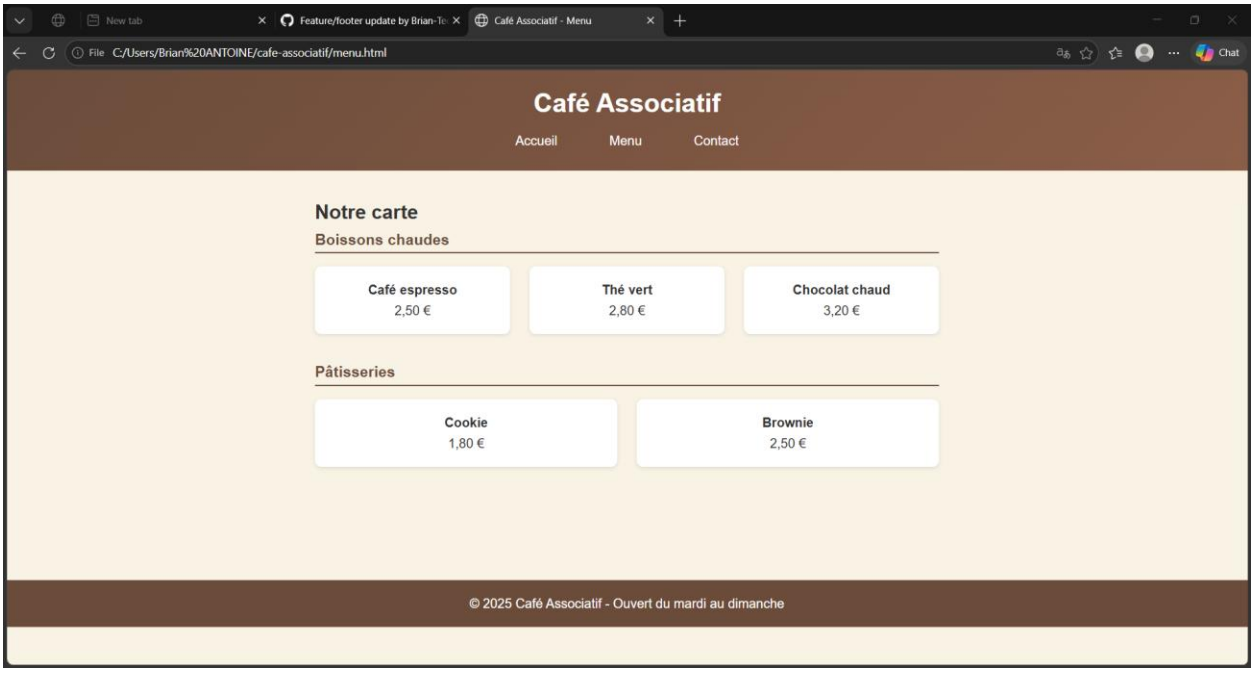
- **Push de la branche résolue vers GitHub**

```
MINGW64:/c/Users/Brian ANTOINE/cafe-associatif

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update|MERGING)
$ git add css/style.css

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update|MERGING)
$ git commit -m " Resolution de conflit: fusion degrade et de la navbar flexbox"
[feature/footer-update 0732a85]  Resolution de conflit: fusion degrade et de la navbar fle
xbox

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$ git push origin feature/footer-update
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 711 bytes | 355.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Rbil1y/cafe-associatif.git
   defe888..0732a85  feature/footer-update -> feature/footer-update

Brian ANTOINE@Bri MINGW64 ~/cafe-associatif (feature/footer-update)
$
```

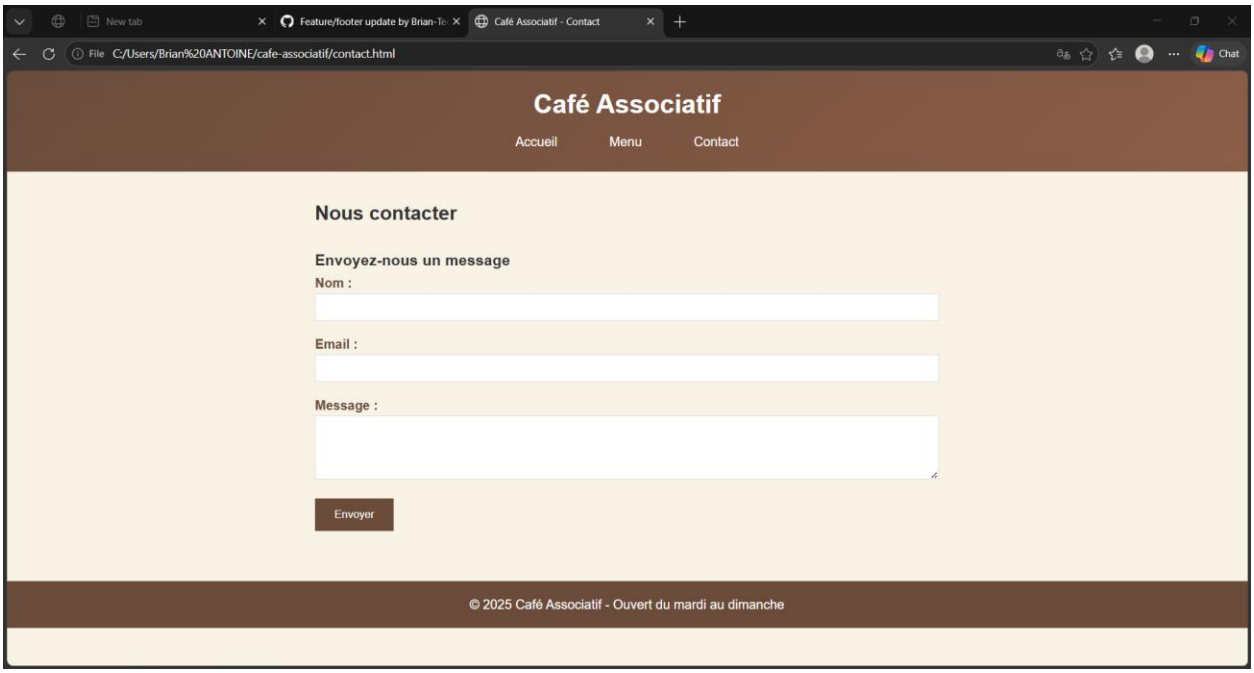
✓ Partie Acceuil



✓ Partie Menu



✓ Partie Contact



Conclusion

Ce TD nous a permis de découvrir concrètement le travail collaboratif avec Git et GitHub. En nous répartissant les rôles : Billy pour l'accueil, Ruth-Jose pour le menu et Brian pour le contact. Nous avons appris à travailler en parallèle sur un même projet sans nous gêner mutuellement grâce aux branches.

Nous avons également pratiqué les Pull Requests pour fusionner notre travail, et surtout vécu une situation réelle de conflit dans le fichier *css/style.css* que nous avons réussi à résoudre manuellement en combinant les modifications des deux développeurs.

La principale difficulté rencontrée a été la résolution du conflit Git, qui était nouvelle pour nous. Mais cette expérience nous a montré que c'est une compétence essentielle dans tout projet en équipe.

Au final, le site est complet, hébergé sur GitHub, et nous repartons avec une maîtrise solide des bases de Git en équipe.